SECTION 1

Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. PROHIBIT: (A) recognize (B) conflict (C) permit (D) comply with (E) shrink from

2. EXTEND: (A) begin (B) advance (C) retract (D) ordain (E) dispose

3. SYNTHETIC: (A) dynamic (B) authentic (C) fruitful (D) open-minded (E) pathetic

4. INGRATITUDE: (A) initiative (B) propensity (C) sincerity (D) appreciation (E) self-denial

5. SPECIFY: (A) anticipate (B) discuss fully (C) return (D) cast doubt upon (E) generalize

6. PROFUSE: (A) ugly (B) tardy (C) meager (D) unlikely (E) disorganized

7. CRAVE: (A) demand silence (B) cheer up (C) disdain (D) cause (E) mar

8. COAGULATE: (A) flow freely (B) disperse quietly (C) complete (D) protect (E) sterilize

9. SLEEPER: (A) obstacle (B) miniature (C) bitter dispute (D) concealed evidence (E) predicted winner

10. INCISIVE: (A) habitual (B) disturbing (C) overbearing (D) obtuse (E) unworthy

11. RESCIND: (A) summarize (B) oppose (C) maximize (D) restore (E) disturb

12. PRODIGIOUS: (A) puny (B) swift (C) distant (D) restless (E) irregular

13. QUALIFIED: (A) underlying (B) disregarded (C) unrestricted (D) predetermined (E) rehabilitated

14. PIQUE: (A) stability (B) servility (C) lethargy (D) delight (E) mercy

15. PULCHRITUDE: (A) dispassion (B) innocence (C) unkindness (D) monotony (E) ugliness

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been —, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. The author includes very little ---- of the dragon: monsters are more fearful if pictured in the imagination.

(A) protection (B) description (C) suspicion (D) defense (E) criticism

17. Human beings are much less creatures of instinct than other animals are; nearly all our behavior patterns are ---- rather than ----.

(A) learned. .inherited (B) predictable. .erratic (C) routine. .diversified (D) innate. .acquired (E) conventional. .arbitrary

18. He angrily ---- Plato as more consistently --- than any other philosopher on all questions involving physical science, politics, ethics, or education.

(A) approved. .debatable (B) defended. .refuted (C) condemned. .wrong (D) quoted. .brilliant (E) derided. .acceptable

19. The business community needs to ---- the role of profits so that their importance will be better understood by all the people who benefit from them.

(A) minimize (B) disclaim (C) elucidate (D) diversify (E) equivocate

20. State ownership of the country's productive resources may unfortunately ---- rather than ameliorate environmental problems.

(A) counteract
(B) expedite
(C) exacerbate
(D) diminish
(E) stigmatize

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1

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Only during the nineteen-thirties did Puerto Rico begin to deviate from colonial patterns and develop her own industry. Though this study investigates mainland cultural influences in the Island's economic development since 1930, attention is not focused on enterprises operated from the mainland. These enterprises have flourished since the beginning of tax exemption, but their operations are not characteristic of Puerto Rican entrepreneurship. The influence of mainland entrepreneurs on the Island and of government programs, however, has not been ignored.

I eschewed any attempt to determine what changes were good or bad for Puerto Rican life. Economic efficiency often comes at the expense of social efficiency or the good life, but I withheld judgment as far as possible.

The two most important forces operating in business are physical and cultural. Geography, climate, cost of transportation, and location of population are all measurable. Culture, however, involves psychological elements. No effort has yet been made to relate these problematic cultural factors to economic development, but by placing them within the economic process, I hope at least to define the problem.

Data was drawn from interviews with established businessmen knowledgeable about the history of business in the Island, interviews with younger entrepreneurs involved in important manufacturing, prior statistics, government documents, and journalists' analyses covering the period of mainland control. The older businessmen were employers of over a hundred workers, though several managed family businesses of five hundred or more. The younger group headed firms employing less than fifty workers.

For comparative purposes, data on entrepreneurial activity in the mainland came from interviews with urban manufacturers and with a panel whose members were selected for diversity in business activity in rural areas.

Questionnaires covered origins of the firm, character of mainland contacts, employee relations, and government relations. Cultural influences were deduced rather than inquired about directly. Mainland business practices were used as an inexact basis for comparison. The inexactness arises from the fact that comparable business activities are larger than those in the Island and operate at a later stage of economic development. Furthermore, the influence of culture on business in the United States has not been thoroughly analyzed. Surprisingly, intensive research in the smaller test area supplied better information than exists for the larger area. This is generally true of investigations into business-culture relationships in larger Western countries.

In spite of these problems, it is possible to formulate hypotheses regarding the comparative character of mainland and Puerto Rican entrepreneurship and the relation of general cultural factors to business. It is first necessary, however, to sketch some aspects of Puerto Rican history that determined the environment for entrepreneurship.

21. The primary purpose of the passage is to

(A) paraphrase a document
(B) provide an introduction
(C) explain the implications of a proposal
(D) refute previously accepted theories
(E) justify a choice of subject matter

22. The author's analysis is based on all of the following EXCEPT

(A) personal interviews
(B) previous studies
(C) newspaper accounts
(D) psychology textbooks
(E) questionnaires

23. The author suggests that cultural factors withstand systematization because they

(A) occur infrequently
(B) change constantly
(C) are unrecognizable
(D) are intangible
(E) are illogical

24. The passage implies that the author wishes to describe the economic changes in Puerto Rico in a tone that is

(A) patriotic (B) humble (C) reflective (D) inspirational (E) objective

25. The author states that the discussion will include the period since 1930 because during that period

(A) mainland influences were completely eliminated
(B) Puerto Rico began to develop locally owned businesses
(C) government programs were created to aid private businessmen
(D) Puerto Rican businesses began to sell more products on the mainland
(E) the Island's international economic relations became more secure

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

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The zoological units (technically, polyps) of every coral mass are of basic coelenterate design, each a simple and usually tiny thimble of translucent tissue fringed, flowerlike, with a corona of waving tentacles, and each loaded with symbiotic algae. The coral polyp surrenders some of its autonomy, affixing its limestone hut snugly to those of its neighbors. What factors or benefits support this type of social cohesion are unknown. In any case, the resulting limestone structure is architecturally distinct for each coral species, of which there are some twenty-five hundred throughout the marine world. A cursory glance into a pool reveals platters, tiers, cerebrally convoluted lumps, fans, a forest of antlerlike branchings—each representing a discrete species. Frustrating, however, to any attempt at quick identification is the fact that a coral colony growing in a quiet tide pool may vary considerably in configuration from that of the identical species located on the buffeted outer reef. Clearly, the form into which the individual stone chambers are welded is affected by the physical character of the environment.

After the normal demise of coral polyps, their calcareous remains persist, and on these grow new polyps, with new limestone thus ever being added to the sun-exposed outer surface of the ancestral structure. In the course of geological time, this amassment of coral gravestones—along with a miscellaneous accumulation of mollusk shells, arthropod and echinoderm husks, foraminifera shells, and the leavings of such calcareous sea plants as Lithothamnion and Porolithon—builds shores, atolls, and islands of staggering bulk. Some reefs grow with only moderate speed, others so rapidly as to render obsolete any navigation charts older than twenty years.

The vast tonnage of limestone underlying every reef's viable upper crust may extend downward even thousands of feet, and this seems puzzling in view of calcareous deposition having occurred only in the sun-illumined shallows of the surface zone. A generally accepted explanation holds that atolls and other coral structures thicken vertically in response to the sea floor's slow subsidence, provided, of course, that reef growth at the surface is fast enough to keep pace with bottom sinkage.

Among destructive elements continually at work on the reef are, paradoxically, those hosts of creatures that themselves find the limestone labyrinth a refuge—the bacteria and protozoa; the sponges, echinoderms, and worms; the mollusks, shrimps, and crabs; the legions of glittery reef fishes—all rubbing, boring, burrowing, and tearing at what the coral masons and their algal collaborators have hewn from sunlight, water, and mineral. And finally, the sea itself, with its eroding tides, violent storms, and crushing surf, contributes to the reef's destruction.

26. This passage is primarily concerned with discussing the

(A) identification of coral
(B) life cycle of coral
(C) nature of coral deposits
(D) origin of islands, atolls, and reefs
(E) organisms that hinder the growth of coral reefs

27. According to the passage, what happens to the remains of dead coral polyps?

(A) They dissolve into the water.
(B) They become a base for new deposits.
(C) They are inhabited by algae.
(D) They are quickly destroyed by erosion.
(E) They sink to the ocean floor.

28. According to the passage, the relationship between coral polyps and algae is one of

(A) parasitism (B) enmity (C) predation
(D) emulation (E) cooperation

29. Which of the following can be inferred about the ocean floor under a very thick coral reef?

I. It has been slowly sinking over a long period of time.
II. It is composed of limestone.
III. It used to be near the surface of the ocean.

(A) I only (B) II only (C) I and II only
(D) I and III only (E) I, II, and III

30. The style of the passage can best be described as

(A) argumentative (B) rhetorical
(C) contemplative (D) expository
(E) derogatory

GO ON TO THE NEXT PAGE

1

Select the word or set of words that best completes each of the following sentences.

31. Susan did not resent the arduous work, for she believed that every ---- that demands thought, attention, and independent judgment ---- the quality of daily life.

(A) task. .heightens (B) profession. .belittles
(C) hobby. .undercuts (D) folly. .exalts
(E) diversion. .disrupts

32. The Sybarites of ancient times were not only physically ---- but also intellectually ----; they had others do everything for them, even their thinking.

(A) active. .independent (B) slothful. .energetic
(C) spent. .brilliant (D) aggressive. .aloof
(E) indolent. .lazy

33. Modern physicists are inclined to believe in the validity of general relativity for ---- reasons, because it is mathematically so elegant and philosophically so ----.

(A) practical. .intricate
(B) technical. .problematical
(C) aesthetic. .satisfying
(D) misguided. .useful
(E) irrational. .verifiable

34. The study of history leads one away from the extremes of belief, thus allowing one to develop conviction without dogmatism, devotion without ----.

(A) comprehension
(B) realism
(C) moderation
(D) fanaticism
(E) skepticism

35. What makes the modern period in art ---- is that this time, in most categories, the older ---- has not been replaced by something equally substantial, accessible, and satisfying.

(A) exacting. .efficiency
(B) ideal. .illusiveness
(C) unique. .ignorance
(D) unprecedented. .reality
(E) worthless. .depravity

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. CAPTAIN : CREW ::
(A) student : faculty (B) mascot : team
(C) defendant : jury (D) chairperson : committee
(E) representative : senate

37. HOMESTRETCH : RACE ::
(A) source : essay (B) gallery : play
(C) finale : opera (D) applause : oration
(E) prelude : concerto

38. ROBBERY : THIEF :: (A) jewelry : burglar
(B) forgery : counterfeiter (C) hostage : kidnaper
(D) sabotage : plunderer (E) capture : fugitive

39. MENAGERIE : ANIMALS :: (A) circus : acrobats
(B) gallery : paintings (C) sachet : powders
(D) archives : events (E) factory : furniture

40. PILGRIM : PIETY :: (A) explorer : curiosity
(B) miser : poverty (C) gambler : winner
(D) knight : beauty (E) monk : loneliness

41. DIPLOMA : GRADUATION :: (A) title : land
(B) supplement : will (C) license : hunter
(D) deed : purchase (E) summons : court

42. FURROW : PLOW :: (A) field : grain
(B) wake : vessel (C) arrow : bow
(D) bale : twine (E) promontory : shore

43. MICROBE : ORGANISM :: (A) microcosm : world
(B) microscope : cell (C) microfilm : camera
(D) animal : vegetable (E) vitamin : protein

44. INCESSANT : RESPITE ::
(A) static : progress
(B) strategic : crisis
(C) inaudible : silence
(D) eventual : possibility
(E) catastrophic : accident

45. LAMPOON : RIDICULE :: (A) dirge : frivolity
(B) treatise : diligence (C) diatribe : mediocrity
(D) panegyric : praise (E) encore : entertainment

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

25 QUESTIONS

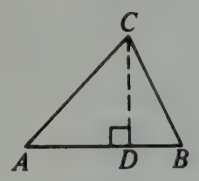
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

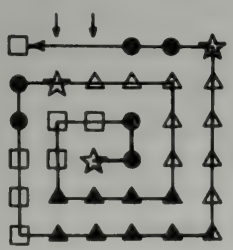
Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. Three dogs weigh 6, 10, and 12 kilograms, respectively. The weight of the lightest dog is how much less than the average (arithmetic mean) weight of the three dogs?

   (A) 6 kg (B) 4 kg (C) $3\frac{2}{3}$ kg

   (D) $3\frac{1}{3}$ kg (E) $2\frac{2}{3}$ kg



2. In the figure above, the pattern is repeated every 15 symbols. Which of the following, when placed below the arrows in the design above, will continue the pattern of the design?

   (A) □ □ (B) ● ● (C) ▲ ▲

   (D) ☆ △ (E) △ ☆

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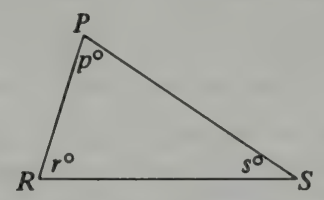
3. 50 per cent of 50 per cent of 1 is

   (A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) 1 (E) 2

4. From which of the following statements taken separately or together can it be determined that x is greater than y ?

   I. $x > 17$
   II. $y < 10$

   (A) I alone but not II
   (B) II alone but not I
   (C) I and II taken together but neither taken alone
   (D) Both I alone and II alone
   (E) Neither I nor II nor both



Note: Figure not drawn to scale.

5. In $\triangle RPS$ above, if $RP < PS$, which of the following is true?

   (A) $s > p$ (B) $r > p$ (C) $r < s$
   (D) $r = s$ (E) $r > s$

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6. If $y = \frac{1}{xz}$, what is the value of x when $y = \frac{1}{6}$ and $z = 2$?

(A) $\frac{1}{12}$

(B) $\frac{1}{8}$

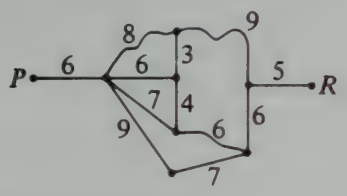
(C) $\frac{1}{3}$

(D) 3

(E) 12

7. If xy is positive, which of the following CANNOT be true about x and y ?

(A) $x < y < 0$
(B) $y < x < 0$
(C) $x < 0 < y$
(D) $0 < x < y$
(E) $0 < y < x$

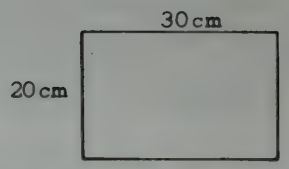


8. If the road distances between any two points are as indicated on the map above, what is the shortest road distance from P to R ?

(A) 27 (B) 28 (C) 29 (D) 30 (E) 33

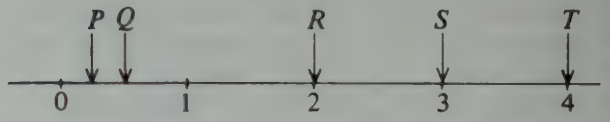
9. A number m is doubled, the result is then increased by 3, and this result is multiplied by 5. Which of the following expressions corresponds to the statement above concerning m ?

(A) $5(2m + 3)$
(B) $2m + 3(5)$
(C) $15(m + 2)$
(D) $5[3(m + 2)]$
(E) $5[(m + 2) + 3]$



10. In the scale drawing of the rectangular dining room shown above, the scale used was 5 cm = 1 meter. What is the actual area, in square meters, of the room?

(A) 10 (B) 12 (C) 24 (D) 120 (E) 250

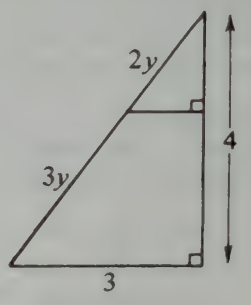


11. Of the following, the closest approximation to the product of the five numbers indicated by arrows on the number line above is

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) 2 (D) 3 (E) 6

12. For how many integers x is $3 < 2x < 5$?

(A) None (B) One (C) Two
(D) Three (E) Infinitely many



13. In the figure above, $y =$

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

GO ON TO THE NEXT PAGE

14. The sum of the first n positive integers $1+2+3+\ldots+n$ is $\frac{n(n+1)}{2}$. The sum of the first 25 positive integers is

(A) 150
(B) 300
(C) 325
(D) 351
(E) 650

15. What is the sum of the areas of two squares with sides of 1 and 3, respectively?

(A) 3
(B) 4
(C) 9
(D) 10
(E) 16

16. $\sqrt{1-\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}\right)}=$

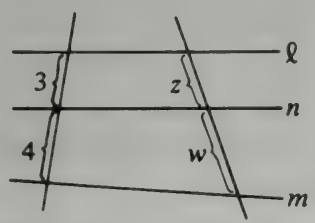
(A) $\frac{1}{2\sqrt{2}}$

(B) $\frac{1}{4}$

(C) $\frac{1}{4\sqrt{2}}$

(D) $\frac{1}{8}$

(E) $\frac{1}{16}$



Note: Figure not drawn to scale.

17. In the figure above line ℓ is parallel to line n, but line n is not parallel to line m. If $z+w=5$, what is the ratio of z to w?

(A) 1:3 (B) 2:3 (C) 3:4 (D) 4:3
(E) It cannot be determined from the information given.

18. In one month, a fleet of trucks used 2,000 liters of gasoline at \$0.25 per liter. If the total distance traveled by the fleet during that time was 4,000 kilometers, the gasoline cost per kilometer was approximately

(A) \$0.050
(B) \$0.0625
(C) \$0.0833
(D) \$0.125
(E) \$0.222

19. In a certain cave with a ceiling 5 meters high, a stalactite begins to grow downward from the ceiling at the rate of 0.004 centimeters per year. At the same instant a stalagmite begins to grow upward directly beneath the stalactite at the rate of 0.006 centimeters per year. At these rates how many years would it take for this stalactite and this stalagmite to touch? (1 meter = 100 centimeters)

(A) 500
(B) 2,000
(C) 20,000
(D) 25,000
(E) 50,000

20. The sum of three consecutive odd integers is N. In terms of N, what is the sum of the next three consecutive even integers that follow the greatest of the three odd integers?

(A) $N+6$
(B) $N+12$
(C) $N+15$
(D) $N+18$
(E) It cannot be determined from the information given.

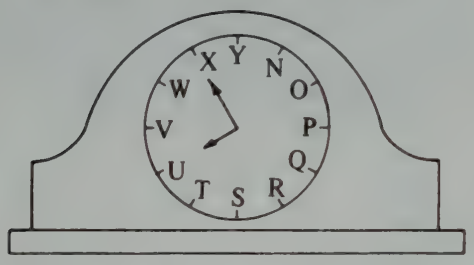
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21. A cube has a volume of 72. If it is divided into 8 equal cubes, the ratio of an edge of a smaller cube to an edge of the original cube is

(A) 1 : 2
(B) 1 : 3
(C) $1 : 3\sqrt{2}$
(D) 2 : 9
(E) 1 : 9

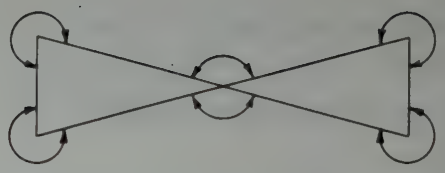
22. If $m^2 = 16$ and $n^2 = 36$, then the difference between the greatest possible value of $m - n$ and the least possible value of $m - n$ is

(A) 20 (B) 10 (C) 4 (D) 2 (E) −2



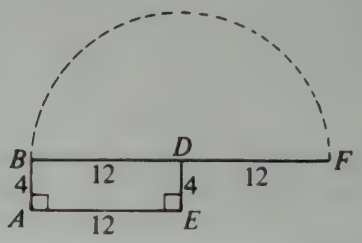
23. The figure above shows a twelve-hour clock marked with letters representing the numbers 1 through 12 in their customary arrangement on its face. Which of the following is a correct representation in these letters of the time indicated by the hands?

(A) X minutes of U
(B) X minutes after T
(C) N minutes of U
(D) R minutes of U
(E) 5X minutes after U



24. What is the sum of the degree measures of all the exterior angles indicated by arrows in the figure above?

(A) 360 (B) 720 (C) 1,080
(D) 1,440 (E) 1,800



25. If, in the figure above, arc BF is a semicircle having radius 12 and C is a point, not shown, on arc BF, which of the following gives all possible values of the perimeter P of pentagon $ABCDE$?

(A) $32 < P < 56$
(B) $32 < P < 44$
(C) $32 < P < 32 + 12\sqrt{2}$
(D) $P = 44$
(E) $P = 32$

S T O P

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DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 3

Time—30 minutes

40 QUESTIONS

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Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. BLUNDER: (A) reveal (B) dash (C) move gracefully (D) change radically (E) accept unquestioningly

2. DISTINGUISHED: (A) temperamental (B) inaccurate (C) commonplace (D) impulsive (E) uncomfortable

3. SUMMARY: (A) difficult question (B) extended account (C) important statement (D) elegant style (E) novel thought

4. CORRUPTION: (A) understanding (B) intervention (C) abstention (D) flexibility (E) purity

5. TEMPERATE: (A) extreme (B) suspect (C) distant (D) independent (E) distasteful

6. UTTER: (A) leave unspoken (B) deny angrily (C) remain unmoved (D) let alone (E) give generously

7. PROVINCIAL: (A) unlucky (B) minor (C) sophisticated (D) average (E) permanent

8. VIRULENCE: (A) fragrance (B) corpulence (C) cleanliness (D) innocuousness (E) ignorance

9. GAMELY: (A) awkwardly (B) seriously (C) without caution (D) without courage (E) without skill

10. PEJORATIVE: (A) approbatory (B) hypothetical (C) vicarious (D) ambiguous (E) tentative

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Nutritionists declare that the mineral selenium, despite its toxic aspects, is ---- to life, even though it is needed in extremely small quantities.

(A) destructive (B) insignificant (C) essential (D) extraneous (E) vulnerable

12. Although much work has gone into the book, it lacks sufficient balance and depth to qualify as a ---- explanation of Cuban economic development.

(A) partisan (B) comprehensive (C) fictitious (D) fragmentary (E) gratuitous

13. It would be ---- for any serious candidate to ---- such an influential constituency.

(A) profitable. .snub (B) illogical. .appease (C) foolish. .offend (D) selfish. .overrate (E) immaterial. .recognize

14. In spite of the ---- of the speech, most of the audience was ---- well before the speaker had finished.

(A) importance. .interested (B) pedantry. .bored (C) excellence. .stirred (D) brevity. .asleep (E) reliability. .convinced

15. Because disuse ---- the intelligence as surely as it withers muscles, grave consequences await humanity if the stimuli for a full utilization of mental powers are diminished further.

(A) whets (B) deters (C) augments (D) distends (E) atrophies

GO ON TO THE NEXT PAGE

3

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. BOOK : PAGES :: (A) question : answers
(B) lighthouse : ships (C) sentence : words
(D) ring : fingers (E) ocean : tides

17. FADE : DISAPPEAR :: (A) sink : swim
(B) wink : smile (C) decelerate : stop
(D) scrub : stain (E) destroy : dismantle

18. IMMEDIATELY : DELAY ::
(A) voluntarily : motive (B) urgently : aid
(C) continuously : effort (D) flawlessly : error
(E) accidentally : injury

19. HORTICULTURE : PLANTS :: (A) printing : ink
(B) geography : planets (C) biology : texts
(D) geology : ruins (E) forestry : trees

20. WINCE : PAIN :: (A) forget : confidence
(B) tremble : fright (C) grovel : embarrassment
(D) glower : anguish (E) growl : delight

21. UPSTREAM : SOURCE :: (A) upwind : vane
(B) uphill : peak (C) upright : foot
(D) upward : angle (E) upstage : theater

22. ARABLE : FARMERS :: (A) habitable : occupants
(B) curable : patients (C) abominable : consumers
(D) legible : writers (E) inaccessible : travelers

23. HACKNEYED : PHRASE ::
(A) obscure : text
(B) unpublished : manuscript
(C) amateurish : profession
(D) frequent : punctuality
(E) threadbare : clothing

24.

This question was not counted in computing scores.

25. DICTUM : DOGMATIC :: (A) mandate : idealistic
(B) jargon : explicit (C) epigram : candid
(D) riddle : cryptic (E) retort : symbolic

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Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

The last time Ivan had tried to do it, his method had been, in the doctor's words, a masterpiece of gruesome inventiveness; he would have succeeded, had not an envious fellow patient thought he was learning to fly—and stopped him. What he really wanted to do was to tear a hole in his world and escape.

The system of his delusions had been the subject of an elaborate paper in a scientific monthly; "referential mania," Herman Brink had called it. In these very special cases the patient imagines that everything happening around him is a veiled reference to his personality and existence. Phenomenal nature shadows him wherever he goes. Clouds in the staring sky transmit to one another, by means of slow signs, incredibly detailed information regarding him. His inmost thoughts are discussed at night by darkly gesticulating trees. He must be always on his guard and devote every minute and module of life to the decoding of the undulation of things. The very air he exhales is indexed and filed away.

26. In the first paragraph Ivan is described as having unsuccessfully attempted to

(A) destroy the prison cell
(B) describe his affliction
(C) establish dominance over the other patients
(D) escape the asylum by airlift
(E) take his own life

27. According to the passage, Ivan's mental representation of the world is a

(A) naïve, harmless illusion
(B) construct without rational organization
(C) system making no allowance for individual worth
(D) structure of integrated, terrifying delusions
(E) complex of notions resembling those of primitive men

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3

John Philip Sousa was no Beethoven. Nevertheless, he was Sousa. When you say "a Sousa march," the phrase means something pretty definite to almost anyone who hears you. Nobody asks, "Which Sousa march?" It does not matter. Any one of them bears the imprint of a vigorous, clear-cut, decidedly original musical personality. They are not "festival" marches, or any other concert variant of the original form. They are intensely practical. Sousa started as a navy bandmaster and did most of his work in the open air and in motion. The marches he wrote, first for the Marine Band and later for his own, were intended to set the pace for marching men.

They have a deceptive simplicity, those Sousa marches. Their tunes are so uncomplicated, so easy to catch, so essentially spontaneous and melodic, that one can easily underrate them. Simple as they may be, they are Sousa's tunes and no one else's. It took only a minor grade of inspiration to write them, perhaps. It was, nonetheless, genuine inspiration.

We do rightly, of course, to judge people by their reach as well as their grasp. It is only fitting to admire Beethoven and Wagner for their pretensions as well as for their achievements. They dared more than others. If they won greater glory, they also risked a more disastrous failure. Yet I think it is not always necessary to be technically "great" in order to be immortal. The giants of art stir our hearts and souls and imaginations. Sousa stirs only our feet. Nevertheless, he does stir them.

28. The primary purpose of the passage appears to be to

(A) set forth a new definition of artistic greatness
(B) defend the worth of the music of John Philip Sousa
(C) emphasize the practical importance of music
(D) explain the relationship between inspiration and immortality
(E) compare the music of Sousa with that of Beethoven

29. The author would be most likely to use which of the following terms in describing the music of John Philip Sousa?

(A) Memorable (B) Intricate (C) Overrated (D) Soul-stirring (E) Pretentious

30. Sousa's work is "practical" (line 10) in the sense that it is

(A) often imitated
(B) easily remembered
(C) readily learned
(D) used for actual military marching
(E) the high point of military concerts

31. Which of the following is the best interpretation of lines 23-24 ?

(A) A person's past generally determines the success that the person will have in the future.
(B) A person who is sincere is as worthy as a person who is successful.
(C) It is wise to consider people's goals as well as their deeds.
(D) It is necessary to applaud the humble as well as the proud.
(E) It is better to praise talent than ambition.

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In published works, drawings and photographs of hailstones are often accompanied by measurements of size and sometimes by data on weight, but not by data on density. The earliest more or less reliable values for density were those obtained by List. These measurements, however, were not made on freshly fallen hailstones but on ones that had been kept before and during measurement at a temperature below 0°C. This fact is of decisive importance, as was established only later, because the hailstones normally contain liquid water and thus exhibit a density greater than that of ice.

It is also important to indicate and classify the forms of hailstones, because their forms are responsible for their aerodynamic behavior when falling and because their forms should permit conclusions to be drawn retrospectively about the conditions in which the hailstones originated. But because different growth phases can succeed one another in the development of these ice particles, it is necessary to learn all the different forms through which a hailstone passes during its growth. For this purpose, cross sections must be prepared in order to gain a picture of the inner structure.

Here the methods devised by M. de Quevrain for research on snow proved to be invaluable. By means of a special circular saw, ice sections 0.2 millimeter or more thick are cut out of the hailstones. A variety of such thin sections can be prepared from any given hailstone. When light penetrates these sections, the arrangement of the air pockets becomes visible, and the forms at different stages of growth may be inferred wherever the air bubbles are arranged in onionskin-like layers. From the symmetry of the sections, it is also possible to recognize the oldest part of the hailstone.

32. The author implies that List's data on the density of hailstones do not describe the density of freshly fallen hailstones because

(A) the evaporation of hailstones during storage will inevitably alter their density
(B) normally a hailstone has a temperature many degrees above 0°C
(C) the hailstones used in his research consisted entirely of ice
(D) his research was a pioneering effort and therefore his data were only "more or less" reliable
(E) it was learned only later that he was dealing with liquid water

33. The author indicates that the value of cross-sectioning techniques lies in the experimenter's ability to use them to

(A) determine the liquid water content of the hailstones he is observing
(B) reconstruct the aerodynamic properties exhibited by the hailstone in falling
(C) determine the characteristics of the most recent layer which was formed
(D) calculate the density of the hailstone
(E) infer the characteristics of the environment in which formation took place

34. Which of the following facts provides the best evidence for the author's statement about the comparative density of ice and liquid water?

(A) Glaciers move very slowly even on quite steep slopes where liquid water runs rapidly.
(B) Icebergs may be seen drifting quite far south in the Atlantic.
(C) Ice on the wings of an aircraft may make flight impossible, but rainstorms seldom interfere.
(D) It is possible to walk across a frozen stream in winter, but it is not possible when the stream has thawed.
(E) Water appears on the surface of a skating rink on a spring day.

35. On the basis of the passage, which of the following substances seem(s) to be present in the average freshly fallen hailstone?

I. Ice
II. Air
III. Liquid water

(A) I only
(B) III only
(C) I and II only
(D) I and III only
(E) I, II, and III

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Alice, as you will recall, was a level-headed sensible girl, with the role of straight man for the remarkable characters in whose company she found herself. She asked the honest questions. They gave the brilliant replies. You will also recall that in her brief visits she was never quite integrated into Wonderland and the Looking-Glass world—in that universe but not of it. If she had been destined to stay permanently with her new playmates, something would have been necessary to correct this state of social maladjustment. Alice had a certain lack of insight, and all hands might have done with some psychotherapy.

I suppose it is because of its history, the course of its earlier development and its origins, that psychology is not yet fully integrated with the realm of biology though, inescapably and permanently, it is now in that realm. We do not communicate well enough: not as geneticist communicates with botanist, or physiologist with biochemist. A psychologist, even a comparative psychologist, could not step into a job in zoology as a physiologist might into pharmacology or neuroanatomy. The clinical neurologist complains that psychologists are complicating the problem of aphasia; the neurosurgeon does not understand what the objections are to localizing a stuff called consciousness or memory or something else in this part of the brain or that. For their part, psychologists too often fail to keep themselves informed about what goes on in the neurological field and, in defense of such ignorance, too often deny that it has any relevance for their work—a position so preposterous and indefensible that it is hard to attack.

There are no tougher problems than ours, nor more urgent ones. If it were possible to communicate better, we might increase the rate of progress in solving them. I do not hold much with team research, except in the applied field ("team research" here means unified control of what the individual investigator does, either by a director of research or by committee). I am not talking about this, but about the stimulation to new ideas, the criticism and guidance that is possible when workers with different backgrounds and skills effectively understand each other's language and modes of thought.

It may not be bias for me to argue that for such collaboration the electrophysiologist and neurochemist must do more homework in psychology. The converse holds as well, and I have put in a fair amount of effort persuading psychologists to do more boning up on the biological bases of psychology.

I do not think it fair to say that a camel will go through the eye of a needle sooner than one could make a psychologist out of an electrophysiologist or an anatomist. That would be inaccurate as well as unfair. I know workers in other biological disciplines who really understand modern psychology, its whys and wherefores. But such understanding demands as much effort as for a psychologist learning the anatomy of the reticular formation or the functioning of the pituitary. My own estimate frankly is that it demands more.

36. The author's attitude toward his subject matter is that of a

(A) psychologist applauding biologists
(B) biologist criticizing psychologists
(C) biologist persuading psychologists to study biology
(D) biologist describing psychology's dependence on biology
(E) psychologist urging greater contact between psychology and biology

37. The author implies that when psychology was first developed it was

(A) more respected than it is now
(B) closely allied with genetics
(C) less involved with team research
(D) not a unique and independent science
(E) not included among the biological sciences

38. With which of the following statements would the author be LEAST likely to agree?

(A) Psychologists should become more familiar with zoology.
(B) There should be greater control over the individual research done in the biological sciences.
(C) Physiologists are generally well acquainted with pharmacology and neuroanatomy.
(D) It is possible for an electrophysiologist to comprehend psychology well.
(E) There is an effective interchange of ideas between geneticists and botanists.

39. In the last sentence of the passage, the word "it" refers to the

(A) comprehension of psychology by workers in other biological disciplines
(B) ability of a psychologist to learn the anatomy of the reticular formation
(C) realization that a camel will not go through the eye of a needle
(D) effort required to make a psychologist out of an anatomist
(E) communication between psychologists and scientists in other fields

40. It can be inferred that the clinical neurologist regards aphasia as a

(A) psychological phenomenon to be treated by psychologists
(B) physiological phenomenon to be treated by neurologists
(C) psychological-physiological phenomenon to be treated by a team of psychologists and neurologists
(D) disorder of the memory aspects of the consciousness
(E) neurological problem that has been neglected by psychologists

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

35 QUESTIONS

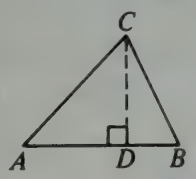
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $0 < a < b < c$, which of the following products is greatest?

   (A) $a \times a$ (B) $a \times c$ (C) $b \times b$
   (D) $b \times c$ (E) $c \times c$

2. If $\frac{2}{3}x = 1$, then $\frac{1}{3}x =$

   (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$ (E) 2

3. In the mid-Atlantic, a ship is traveling in a direction 10° east of north. The captain swings the ship 15° to the left. In what new direction is the ship traveling?

   (A) 25° east of north
   (B) 25° west of north
   (C) 10° west of north
   (D) 5° west of north
   (E) 5° east of north

4. If the "center" of a number is defined to be $\frac{1}{2}$ of the number, then what number is the "center" of itself?

   (A) −1 (B) 0 (C) 1 (D) 2 (E) 10

5. How many bottles of soda at 35 cents each can be purchased with an amount equal to the cost of 14 bottles at 90 cents each?

   (A) 18
   (B) 24
   (C) 30
   (D) 36
   (E) 42

6. If $5x = 3$ and $3y - 5x = 9$, what is the value of y ?

   (A) 2
   (B) 3
   (C) 4
   (D) 5
   (E) It cannot be determined from the information given.

7. The 1st term of a sequence is 1. Each term after the 1st is found by adding 3 to the previous term and then multiplying this sum by 3. Thus 12 is the 2nd term. Which of the following describes the terms of this sequence?

   (A) All the terms are odd.
   (B) All the terms are even.
   (C) The terms follow the pattern: odd, even, odd, even, etc.
   (D) The terms follow the pattern: odd, odd, even, even, odd, odd, even, even, etc.
   (E) The terms follow the pattern: odd, even, even, odd, even, even, etc.

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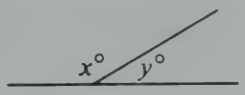
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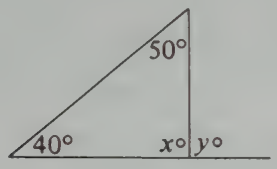
Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

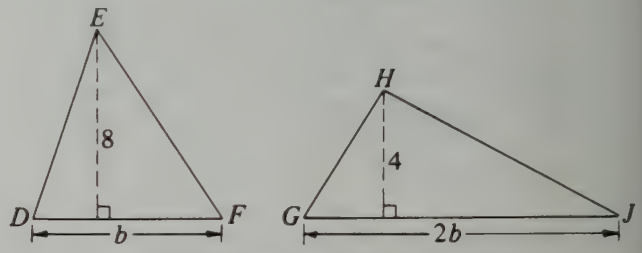
A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	$\frac{1}{4}$ dozen	4
9.	 x	y
10.	There are 51 books on a shelf. Two-thirds of the books have hard covers and the rest are paperbacks. Number of paperbacks	18
	$w + x > y + z$	
11.	w	z

	Column A	Column B
	$a > 0$	
12.	$2a \times \frac{a}{4} \times 8a$	$\frac{a}{2} \times 4a \times \frac{a}{8}$
13.	$\frac{4}{7}$ of 7	$\frac{7}{4}$ of 4
14.	$27 \times (54 + 39)$	$(27 \times 54) + 39$
15.	z	$\frac{2z + 1}{2}$
	$x > 1$	
16.	$\frac{x + x + x}{x \cdot x}$	$\frac{3}{x^2}$
17.	 Note: Figures not drawn to scale. Area of $\triangle DEF$	Area of $\triangle GHJ$

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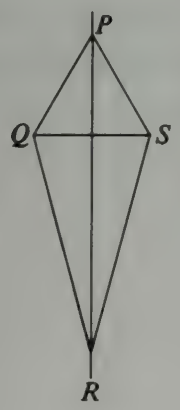
SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

	Column A	Column B
18.	$0.5 + 0.4 + 0.3$	0.3×0.4

$x \neq \pm 2$

19.	$\frac{x^2 + 4x + 4}{x + 2}$	$\frac{x^2 - 4}{x - 2}$



Area $\triangle PQR <$ area $\triangle PSR$ and $QS \perp PR$.

20.	$PQ + QR$	$PS + SR$

* represents one of the operations $+, -, \times$, or $\div$ and $x * x = 0$ for all numbers x.

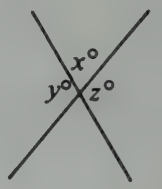
21.	$x * y$	$y * x$

$\frac{p}{q} = 1$ and $\frac{q}{p} = 1$

22.	p	1

23.	Ratio of $\frac{1}{2}$ to $\frac{1}{3}$	Ratio of $\frac{2}{3}$ to $\frac{1}{2}$

	Column A	Column B



$x + y + z = 4x$

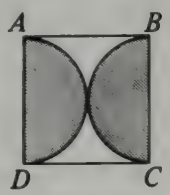
24.	$\frac{3}{2}x$	y

x is a positive number.

25.	x	$\frac{1}{x}$

Ann has 6 more marbles than Nancy, Nancy has 3 more marbles than Joe, and Joe has 4 more marbles than Pete.

26.	The least number of marbles that must change hands if each is to have an equal number of marbles	8



$ABCD$ is a square and the two shaded regions are semicircles.

27.	The ratio of the total shaded area to the area of the square	$\frac{\pi}{4}$

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5

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

28. At some point in the season, five teams have the following records:

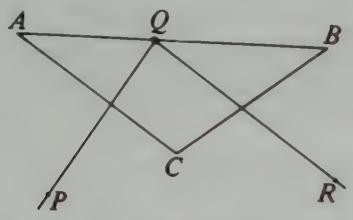
	Games Won	Games Lost
Team 1	3	1
Team 2	4	2
Team 3	6	3
Team 4	5	2
Team 5	3	2

If there were no ties, which team has won the greatest per cent of its games?

(A) Team 1
(B) Team 2
(C) Team 3
(D) Team 4
(E) Team 5

29. $5 \times 10^{-3} \times 6 \times 10^{2} \times 2 \times 10^{-1} =$

(A) 0.000060 (B) 0.060 (C) 0.60
(D) 6.0 (E) 6,000,000

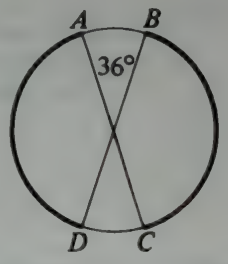


30. In the figure above, if $\angle AQR = 145°$ and $\angle BQP = 122°$, then the degree measure of $\angle PQR =$

(A) 87 (B) 88 (C) 89 (D) 90 (E) 91

31. If the odometer of an automobile registers 62,222 miles, what is the LEAST number of miles that the automobile must travel before the odometer again shows four of the five digits the same?

(A) 99
(B) 444
(C) 555
(D) 999
(E) 1,111

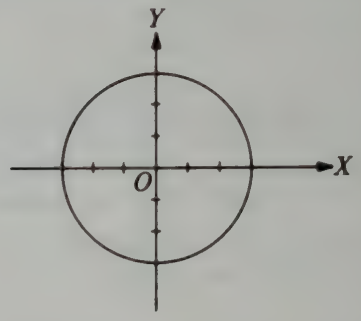


32. In the circle above, AC and BD are diameters of length 10. The sum of the lengths of the darkened arcs is

(A) 4π (B) 8π (C) 9π (D) 10π (E) 16π

33. A carpenter used $\frac{1}{3}$ of his lumber for one project and $\frac{3}{5}$ of what was left for another project. If he had 30 units of lumber to start with, how many units did he have left after the two projects?

(A) 8
(B) 6
(C) 4
(D) 3
(E) 2



34. The circle above has center at the origin and radius 3. How many ordered pairs (x, y) satisfy the conditions that x and y are integers and (x, y) is inside the circle shown?

(A) 9 (B) 13 (C) 16 (D) 20 (E) 25

35. For all numbers x and y, let $x \diamondsuit y$ be defined by the equation $x \diamondsuit y = (x - y)^2 + (x + y)^2$. What is the value of $\sqrt{28} \diamondsuit \sqrt{23}$?

(A) 56 (B) 100 (C) 100.1
(D) 102 (E) 105.25

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test Form Code 2Y

VERBAL		MATHEMATICAL	
Section 1	Section 3	Section 2	Section 5
1. C	1. C	1. D	1. E
2. C	2. C	2. A	2. B
3. B	3. B	3. B	3. D
4. D	4. E	4. C	4. B
5. E	5. A	5. E	5. D
6. C	6. A	6. D	6. C
7. C	7. C	7. C	7. C
8. A	8. D	8. B	*8. B
9. E	9. D	9. A	*9. C
10. D	10. A	10. C	*10. B
11. D	11. C	11. D	*11. D
12. A	12. B	12. B	*12. A
13. C	13. C	13. A	*13. B
14. D	14. D	14. C	*14. A
15. E	15. E	15. D	*15. B
16. B	16. C	16. B	*16. A
17. A	17. C	17. E	*17. C
18. C	18. D	18. D	*18. A
19. C	19. E	19. E	*19. C
20. C	20. B	20. C	*20. B
21. B	21. B	21. A	*21. D
22. D	22. A	22. A	*22. D
23. D	23. E	23. D	*23. A
24. E	24. †	24. D	*24. C
25. B	25. D	25. A	*25. D
26. C	26. E		*26. C
27. B	27. D		*27. C
28. E	28. B		28. A
29. D	29. A		29. C
30. D	30. D		30. A
31. A	31. C		31. B
32. E	32. C		32. B
33. C	33. E		33. A
34. D	34. B		34. E
35. D	35. E		35. D
36. D	36. E		
37. C	37. E		
38. B	38. B		
39. B	39. A		
40. A	40. B		
41. D			
42. B			
43. A			
44. A			
45. D			

*Indicates four-choice questions. (All of the other questions are five-choice.)
†Not scored

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the first two characters (number and letter) of the form code you marked in item 3 on your answer sheet are the same as the form code printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 3

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 3* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 226. Enter this figure on the worksheet.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 226. Enter this figure on the worksheet.

SAT SCORING WORKSHEET

FORM CODE 2Y

SAT-Verbal Sections

A. Section 1: _____ no. correct − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 3: _____ no. correct − ¼ (_____ no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on the back cover.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 2: _____ no. correct − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ no. correct − ¼ (_____ no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ no. correct − ⅓ (_____ no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on the back cover.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Code 2Y

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
			40	470	600
84	800		39	460	600
83	790		38	450	590
82	780		37	450	580
81	770		36	440	570
80	760		35	430	560
79	750		34	430	550
78	750		33	420	540
77	740		32	410	530
76	730		31	400	520
75	730		30	400	520
74	720		29	390	510
73	710		28	380	500
72	700		27	370	490
71	700		26	370	480
70	690		25	360	470
69	680		24	350	460
68	670		23	350	450
67	670		22	340	450
66	660		21	330	440
65	650		20	320	430
64	650		19	320	420
63	640		18	310	410
62	630		17	300	400
61	620		16	290	390
60	620	800	15	290	380
59	610	790	14	280	370
58	600	780	13	270	370
57	590	770	12	260	360
56	590	760	11	260	350
55	580	750	10	250	340
54	570	740	9	240	330
53	560	730	8	240	320
52	560	720	7	230	310
51	550	710	6	220	300
50	540	700	5	210	300
49	540	690	4	210	290
48	530	680	3	200	280
47	520	670	2	200	270
46	510	660	1	200	260
45	510	650	0	200	250
44	500	640	– 1	200	240
43	490	630	– 2	200	230
42	480	620	– 3	200	220
41	480	610	– 4	200	220
			– 5	200	210
			– 6 or below	200	200